



Article

Evaluating Educational Game Design Through Human–Machine Pair Inspection: Case Studies in Adaptive Learning Environments

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Abstract

Educational games often fail to effectively merge game mechanics with educational goals, lacking adaptive feedback and real-time performance monitoring. This study explores how Human–Computer Interaction principles and adaptive feedback can enhance educational game design to improve learning outcomes and user experience. Four educational games were analyzed using a mixed-methods approach and evaluated through established frameworks, such as the Serious Educational Games Evaluation Framework, the Assessment of Learning and Motivation Software, the Learning Object Evaluation Scale for Students, and Universal Design for Learning guidelines. In addition, a novel Human–Machine Pair Inspection protocol was employed to gather real-time data on adaptive feedback, cognitive load, and interactive behavior. Findings suggest that Human–Machine Pair Inspection-based adaptive mechanisms significantly boost personalized learning, knowledge retention, and student motivation by better aligning games with learning objectives. Although the sample size is small, this research provides practical insights for educators and designers, highlighting the effectiveness of adaptive Game-Based Learning. The study proposes the Human–Machine Pair Inspection methodology as a valuable tool for creating educational games that successfully balance user experience with learning goals, warranting further empirical validation with larger groups.



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1. Introduction

Educational games are powerful tools for boosting student engagement and improving learning outcomes. They play a vital role in transforming traditional learning environments, turning static classrooms into dynamic, interactive spaces [1]. By blending educational content with captivating gameplay, these tools harness interactivity to spark curiosity, keep students engaged, and promote deeper learning. As a result, Game-Based Learning (GBL) has grown in popularity and effectiveness in recent years [2]. GBL provides students with opportunities to tackle complex and abstract concepts through hands-on, experiential activities and interactive engagement. Techniques borrowed from video games—such as achievement badges, leaderboards, and contextual guidance—are essential for maintaining motivation [3] and for creating environments where effort is visibly rewarded [4]. Engaging in tasks like solving challenging problems (e.g., decoding) or breaking down complex,

multi-level problems into simpler ones helps students solidify concepts and reinforce learning objectives, often within an enjoyable context [5].

Although the terms Game-Based Learning (GBL) and gamification are often used interchangeably, the literature draws a clear distinction: GBL involves the use of complete games or serious games where rules, narrative, and mechanics are intrinsically aligned with learning objectives to create immersive, contextualized experiences, whereas gamification applies individual game design elements—such as points, badges, leaderboards, and progress bars—to non-game contexts to boost motivation without delivering the holistic experience of a game. Game-Based Learning is widely recognized as an effective strategy to foster motivation and engagement [6], particularly when embedded within constructivist and socio-cultural paradigms [7]. This distinction is grounded in Deterding et al. [8], who define gamification as the application of “gameful” elements in non-game settings, while critics like Bogost [9] warn that superficial gamification can result in shallow engagement when not paired with meaningful content or narrative. Effective GBL design, therefore, requires embedding learning deeply within game mechanics rather than relying solely on extrinsic reward structures. Additionally, contemporary Science, Technology, Engineering, and Mathematics (STEM) education has increasingly embraced the broader STEAM (Science, Technology, Engineering, Arts, and Mathematics) approach by incorporating the arts to foster creativity, design thinking, and human-centered innovation. Integrating the arts amplifies the capacity of scientific and technical learning experiences to produce multidimensional, empathetic solutions, and recent work shows that STEAM promotes engagement, interdisciplinary understanding, and creative problem-solving in Game-Based Learning environments [10].

However, the success of educational games is not solely dependent on students’ motivation to engage. The ongoing interaction between the player and the game system plays a significant role, particularly when it is supported by timely and adaptive feedback mechanisms [11]. Such feedback not only sustains engagement but also supports learning by helping learners monitor and regulate their progress [12]. In this regard, principles from Human–Computer Interaction (HCI) is highly relevant, as it emphasizes usability, personalization, and the optimization of user experience in interactive systems [13]. Recent advances in adaptive gamification further highlight the importance of tailoring in-game responses to individual learners through dynamic feedback loops and user modeling. Core HCI principles include usability, consistency, feedback responsiveness, error prevention, and personalization—each directly influencing educational game engagement and effectiveness [14]. Games need to react dynamically to what the player does. Features like consistent real-time feedback challenge that scale appropriately, and smart progress tracking works with such features to create a truly personal learning journey [15]. This journey adjusts based on the student’s performance, maintaining an optimal balance between challenge and engagement. Such adaptive systems do not just pull players deeper into the game world (enhancing immersion); they actively help players acquire knowledge and skills by ensuring the game mechanics respect cognitive load principles and align with effective problem-solving approaches [16].

Knowing whether these games work is, naturally, a major focus in education today. Rigorous evaluation is a key, not just to seeing immediate results, but also to understanding the long-term effects these games have on learning [17]. Structured approaches to development, like the traditional waterfall model (Figure 1), enrich quality control from start to finish. Specific evaluation frameworks, such as the Assessment of Learning and Motivation Software (ALMAS), the Learning Object Evaluation Scale for Students (LOES-S), the Serious Educational Games Evaluation Framework (SELEAG), the Educational Context-Based Assessment (ECBA), the GameFlow model, Universal Design for Learning (UDL), and

the Model for the Evaluation of Educational Games (MEEGA+), provide tools to examine usability, player motivation, and learning effectiveness, checking that games achieve their objectives both technically and educationally [18]. These evaluation approaches often highlight the importance of easy-to-use interfaces, engaging narratives, and smart adaptive mechanics in keeping players engaged and improving learning retention. Going a step further, methods focusing on player interaction allow for collecting real-time data about user engagement and thinking processes. This information can then feed back into refining the game's design and tailoring the educational content dynamically. The precise role of HCI principles is to apply them in a way that ensures educational games are flexible, truly responsive to individual learners, and welcoming to all (inclusive); this ultimately serves to bridge the gap between engaging entertainment and effective, structured learning [19].

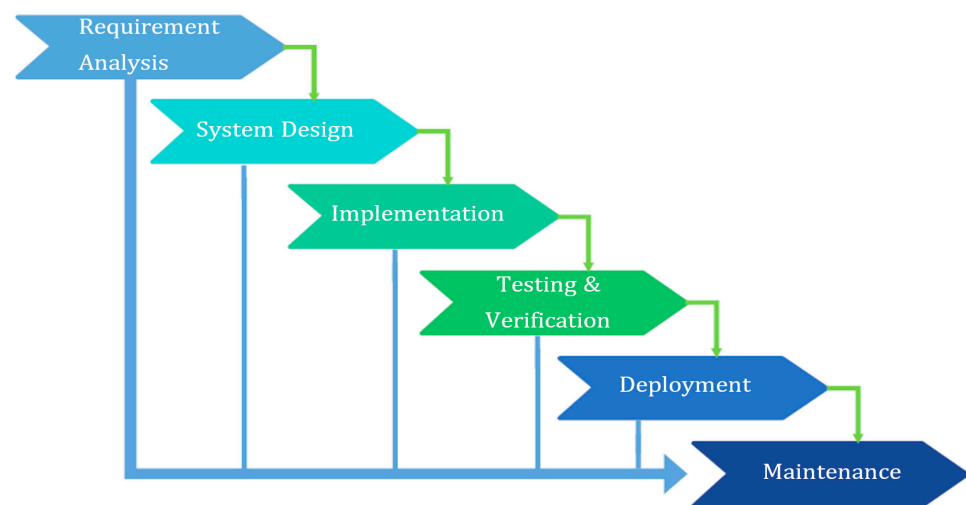


Figure 1. The phases of educational software development according to the waterfall model.

Building on these foundations, this study aims to answer the following research questions:

1. How can game design principles and interactive mechanisms be effectively aligned with educational objectives to enhance student engagement and learning outcomes?
2. What are the advantages and limitations of Human–Machine Pair Inspection (HMPI) in evaluating educational games, and how do these affect learning outcomes and player engagement?
3. In what ways do different evaluation frameworks—such as ALMAS, ECBA, GameFlow, and SELEAG—differ in how they measure the effectiveness of educational games in terms of player motivation, cognitive skill development, and knowledge retention?

To explore these questions, this paper delves into how specific game elements—challenges, puzzles, engaging narratives—can foster crucial skills like critical thinking, teamwork, and problem-solving. The paper also examines the potential of newer technologies like micro:bit, Arduino, and Virtual/Augmented Reality (VR/AR) to create learning environments that are both immersive (‘experiential’) and safe places to experiment.

1.1. Game Design Principles

Grounded in research, educational game design balances engagement and learning value by leveraging STEAM to combine analytical rigor with creativity [20]. Games built around adventure or treasure hunts, for example, can create particularly immersive and meaningful learning journeys [21]. Crafting these games requires carefully linking game mechanics to specific learning goals while encouraging active participation through challenges that push for deeper understanding. Storylines are vital, too; they provide context that boosts engagement and ties game activities to believable situations. Well-designed

games also mirror effective real-world teaching, weaving in practical tasks and interactive methods to boost learning [22]. Logically, evaluation must ascertain the actual impact of these games, confirming that design elements yield meaningful improvements in learning. As exemplified in Section 1.7, the case studies instantiate these STEAM-based, transdisciplinary approaches through narrative, puzzles, and creative challenges.

Within educational games, “Game Design Principles” refers to the core strategies for weaving learning content and teaching approaches into the gameplay itself. Key principles of game design include:

- **Alignment of Mechanics with Learning Objectives:** Ensuring that gameplay directly reinforces the intended learning outcomes.
- **Embedded Assessment:** Integrating evaluation of player learning within the game experience.
- **Adaptive Difficulty:** Adjusting the challenge level based on individual player performance.
- **Meaningful Feedback:** Providing feedback that is both informative and relevant to the learning goals.
- **Motivation and Engagement:** Designing gameplay that sustains player interest and encourages continued learning.
- **Interdisciplinary Integration:** Combining scientific, technical, and artistic perspectives to foster both analytical and creative learning.

1.2. Aligning Game Mechanics with Learning Objectives

A core challenge in educational game design is ensuring every game feature serves the learning objectives. Research backs this up, showing a strong link between well-integrated educational content and engaging mechanics, which ultimately leads to better learning results [23]. Studies also confirm that weaving learning directly into the gameplay helps students acquire and remember knowledge more effectively [24]. Game challenges, mechanics, and rewards should be designed to build skills like problem-solving, critical thinking, and teamwork [25]. For instance, math games might use logic puzzles to sharpen analytical abilities. Clear in-game instructions are also important, so players understand the goals and how to play, letting them focus on the learning itself.

While educational game design is often approached from an instructional or cognitive lens, perspectives from game studies enrich our understanding of how rules, mechanics, and narrative framing shape player engagement and meaning-making. For instance, Bogost [9], Juul [26], and Salen and Zimmerman [27] highlight that games function as semiotic systems—where rules do not merely regulate but express pedagogical intent. Integrating this perspective supports the design of mechanics that not only challenge the player cognitively but also position the learner within an expressive narrative world with educational consequences.

Information and Communication Technologies (ICT) support the creation of engaging adventure and treasure-hunt games. These tools help to build interactive, immersive worlds where learners actively participate and gain hands-on experience. Examples include

- **Virtual Reality (VR):** Immerses students in simulated scenarios, like navigating a rainforest, designing virtual prototypes, or conducting virtual experiments safely [28].
- **Augmented Reality (AR) Games:** Enhances physical spaces with digital overlays [29], allowing activities like identifying virtual plants on a nature walk or solving puzzles using virtual objects in the real world [30].
- **Programming Platforms (Hardware):** Students can combine platforms like Micro:bit and Arduino for complex game setups (e.g., escape rooms with physical locks controlled by Arduino, managed by Micro:bit). In a treasure hunt, devices might communicate to navigate a robot or trigger effects upon finding solutions [31].

- **Gamification Platforms:** Tools like *Minecraft Education Edition* allow exploration through interactive environments, perhaps tasking learners to build functional systems to understand processes [32].
- **Environmental Puzzles:** Can require applying understanding of systems, like simulating food webs, modeling environmental impacts, or using physics to build structures in a virtual treasure hunt. The game *The Legend of the Dead Knight's Spirit* provides a prime example: players needed to apply the Pythagorean Theorem to calculate the rope length required to reach a tower, using the moat width and tower height.

Elements like these are common in immersive adventure and treasure-hunt formats. By using these technologies well, educators can create rich worlds where learning happens through exploration and tackling problems. Location-based AR games, for instance, have successfully boosted student motivation and spatial skills by letting learners interact with virtual elements in their actual surroundings.

1.3. Designing Interactive Challenges for Deeper Learning

Captivating puzzles and challenges are the heart of successful adventure and treasure-hunt games, demanding careful design to ensure real educational value. Research confirms that interactive simulations, for example, significantly boost student engagement and deepen understanding [33]. So, when these challenges are carefully crafted to promote deeper learning, they typically incorporate several essential elements, such as the following:

- **Immediate and constructive feedback:** Providing players with real-time guidance that identifies their weaknesses and areas needing further attention.
- **Scenario-based problems:** Using realistic dilemmas to spur critical thinking and problem-solving [34].
- **Collaborative tasks:** Encouraging teamwork and communication through shared goals, which enhances problem-solving skills.
- **Adaptive difficulty:** Adjusting challenge levels to match individual players' progress and learning style, keeping motivation high [35].

There are games like *Super Turtle Adventures*, which introduces players to Logo programming [36], or *Musical Mastermind*, which cleverly combines music theory with computational thinking [37], clearly demonstrating how complex topics can be integrated into an interactive game and transformed into an enjoyable learning experience. Similarly, the *Digital Little Music Makers* project embodies constructionist principles of “making” to teach music composition, further proving the power of creative, hands-on learning in digital environments [38].

The educational value of these games is further enhanced by high-quality digital learning objects. When games include built-in tools that provide immediate feedback or present real-world problems (such as resource management or understanding recycling systems, as shown in Figure 2), learning becomes much more dynamic and meaningful to students. In *Super Turtle Adventures*, students use Logo to program a turtle robot to sort litter into recycling bins. Moreover, such learning objects are vital as they help students connect theoretical knowledge with practical application. For example, students could practice interpreting various types of data [39], conduct virtual experiments in chemistry or genetics [40]. Customizable environments also help cater to different learning styles. By weaving these interactive challenges and objects into the game's narrative, educational content becomes a natural part of the experience. Mixing disciplines within puzzles, like combining physics and engineering, can also be very effective at maintaining learner interest [41].



Figure 2. Learning to code for a greener planet.

1.4. Crafting Compelling Narratives

A good story is often a key to keeping players engaged and giving context to the learning in educational games. Seamlessly integrating learning goals within the game's narrative tends to enhance player motivation for exploration and learning. For example, a game narrative centered on finding sustainable energy sources can teach environmental concepts within an exciting adventure. Drawing on game studies, narrative framing in educational games should be seen not just as context, but as a mechanism for shaping learner subjectivity and agency. The story becomes a framework through which players interpret challenges, inhabit roles, and internalize pedagogical meaning, making narratives essential components of game-based epistemology.

Effective storylines often use immersive themes and believable challenges, such as

- Sustainability Missions: A scientific expedition to discover renewable energy sources educates players about sustainability while entertaining them.
- Role-Playing Scenarios: Players take on various roles, immersing themselves in real-world challenges.

Narrative framing in educational games should be seen not just as context, but as a mechanism for shaping learner subjectivity and agency. The story becomes a framework through which players interpret challenges, inhabit roles, and internalize pedagogical meaning.

1.5. Research-Backed Design Strategies

Research points to several core strategies for creating impactful educational games:

- Alignment with Learning Objectives: Every game element should support measurable educational goals [42].
- Design and Aesthetics: Visual appeal and overall design significantly influence how usable and credible the game feels [43].
- Usability: The game must be easy to navigate and interact with.
- Engaging Gameplay: Variety in challenges, immediate feedback, and a sense of accomplishment are crucial for motivation [44].
- Appropriate Difficulty: Challenges should adapt to different learners to prevent frustration or boredom; balancing difficulty with adaptive feedback and real-time analytics deepens engagement [45].

- Constructivist/Problem-Based Learning: Games should encourage active learning through exploration, experimentation, and tackling problems.
- Accessibility: Games need to be usable by all learners, including those with disabilities, incorporating features like adjustable difficulty, alternative inputs, or text-to-speech options. Technological accessibility must also account for content formats such as video captions, audio descriptions, and compatibility with screen readers—especially in multimedia-rich educational environments [46].

1.6. Real-World Examples and Resources

Real-world examples highlight the power of these approaches in education:

- *DragonBox*: Uses engaging card games and puzzles to teach algebraic concepts [47].
- *Labster*: Offers interactive simulations of lab experiments across various science fields [48].
- *Game Development Tools*: Platforms like Scratch, Unity, and GameMaker Studio provide user-friendly interfaces and resources, empowering educators and students alike to create their own interactive learning experiences [49].

The principles and strategies discussed above—from aligning game mechanics with learning goals to crafting engaging challenges and narratives—provide a foundation for effective educational game design. To see how these concepts are translated into practice, this section examines four distinct case studies. These examples illustrate different approaches to leveraging gamification and digital tools across various disciplines, highlighting the real-world application and potential impact of thoughtfully designed educational games.

1.7. Case Studies and Applications

Presented here are four case studies of educational games combining gamification and digital technologies to support learning across subjects. The selected titles, (1) *Super Turtle Adventures*, (2) *Musical Mastermind*, (3) *The Legend of the Dead Knight's Spirit*, and (4) *Urban History of Drapetsona*, incorporate thematic or historical narratives alongside puzzle-solving and problem-based learning activities. All four games are rooted in STEAM approaches to foster creativity, cultural awareness, and deeper engagement.

Their thoughtfully developed scenarios are designed to engage learners while promoting both skill development and conceptual understanding. Through the integration of artistic, cultural, or musical dimensions into technical content, the games demonstrate how immersive mechanics can foster transdisciplinary learning. They are carefully aligned with specific educational goals, merging disciplines like programming, music theory, history, and cultural heritage into coherent and meaningful learning experiences.

1.7.1. Super Turtle Adventures

Super Turtle Adventures is a novice programming environment (NPE) designed to introduce students to the Logo programming language through block-based coding puzzles [50]. Beyond developing computational thinking skills, the game also aims to promote environmental awareness. Players guide a virtual turtle through a series of challenges, learning core programming concepts such as movement commands and loops.

In the game *Super Turtle Adventures*, developed by Sarlis et al. (2019) [50], one of the central tasks involves writing code to control a robot that assists a turtle in collecting and recycling waste. This coding component is seamlessly integrated into gameplay, encouraging students to use logic and planning to solve real-world inspired problems. As players proceed, they are required to compose increasingly complex instructions to achieve their recycling goals, making code-writing itself a meaningful part of the problem-solving experience. The game features a visual interface that supports learners as they

progress from simple instructions to more advanced procedural programming. This design encourages the development of logical reasoning and creativity in a format that is both engaging and interactive (see Figure 3). In *Super Turtle Adventures*, students use block-based coding to guide a turtle through puzzles related to recycling. The effectiveness of the game was also assessed using the SELEAG (Serious Educational Games Evaluation Framework), focusing on aspects such as learning outcomes, player motivation, engagement, usability, and overall user experience.



Figure 3. The interface of *Super Turtle Adventures*.

According to the authors, the results of the evaluation pointed to significant gains in programming skills and environmental understanding (see Tables A1–A5 in Appendix A). This evaluation confirmed that *Super Turtle Adventures* effectively boosts students' grasp of programming concepts while keeping them motivated through interactive gameplay tied to real-world environmental themes. To ensure wide access, *Super Turtle Adventures* is available online where students can practice Logo programming anytime, tackling coding puzzles and honing problem-solving skills enjoyably.

1.7.2. Musical Mastermind

Musical Mastermind is an inventive educational tool blending music theory with computational thinking. It offers students an interactive way to decode and program musical sequences using micro:bit v2.0–v2.2 devices. Taking a STEAM-based approach, the game cultivates creativity, logical reasoning, and collaborative learning as students experiment with sound and LED interactions. By combining coding with music composition, *Musical Mastermind* sharpens both technical and artistic skills within a dynamic, engaging experience (Figure 4).



Figure 4. *Musical Mastermind*: Students program a micro:bit to decode the correct musical sequence.

The game's evaluation used the UDL framework, which focuses on accessibility, adaptability, and inclusivity to ensure it meets diverse learner needs. The results of a study by Sarlis et al. (2025) [37] showed that students acquired skills remarkably well, especially in problem-solving, pattern recognition, and computational thinking. A key factor was the high motivation driven by the challenge of cracking and programming musical codes; students showed more interest in exploring music-based puzzles afterward (see Tables A1–A5 in Appendix A). Some students wanted to tackle further coding challenges involving music, underscoring the game's success in building sustained interest and deeper learning across both domains.

1.7.3. *The Legend of the Dead Knight's Spirit*

The *Immersive Gamified Learning in the Old City of Rhodes* project [51] turns a UNESCO World Heritage site into an interactive classroom, using geolocation, QR codes, and story-driven challenges. This engaging setup lets students explore local history while boosting their digital literacy, spatial awareness, and problem-solving abilities. By merging learning with real-world discovery, the project takes a multidisciplinary approach, making history feel more real and relevant through tech-powered storytelling and immersive play (Figure 5).

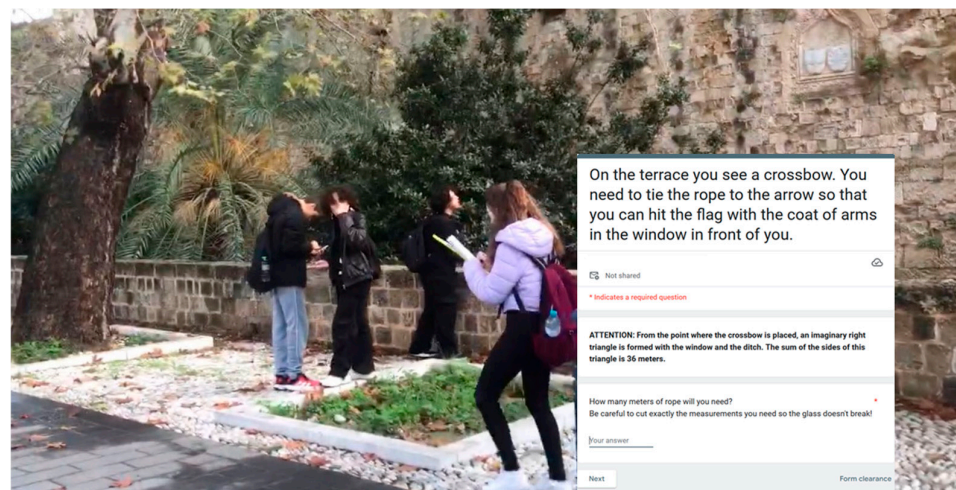


Figure 5. “Riddle of the Rope”: Treasure hunt participants calculate the exact length for the cross-bow challenge.

Evaluation using the LOES-S framework assessed the pedagogical value, content quality, and interactivity of the game. The findings, as revealed by the study of Sarlis et al. (2024) [51] indicated that students effectively learned about the Knights of Rhodes and other historical aspects of the medieval city of Rhodes. The game also motivated students to apply math and physics skills in solving complex puzzles required to advance, which involved calculations, logic, and spatial reasoning grounded in interdisciplinary approaches to create a meaningful learning context (see Tables A1–A5 in Appendix A). Moreover, the evaluation showed that students capably used digital tools, navigating QR codes and Google Maps to find historical spots and access materials on Google Drive. Their skill in interpreting location data and using technology for exploration confirmed the game’s power to improve both historical knowledge and vital 21st-century digital literacy [52].

1.7.4. Educational Cultural Game for Drapetsona’s Industrial Heritage

The *Educational Cultural Game for Drapetsona’s Industrial Heritage* is a point-and-click puzzle adventure drawing players into the city’s industrial past [53]. Using realistic settings and historical storylines, students explore local heritage, tackle interactive challenges, and cultivate logical reasoning, storytelling skills, and civic awareness. By connecting with cultural themes via problem-solving mechanics, the game aims to build a sense of place and appreciation for industrial history, making the learning experience relevant and engaging. It features mini challenges testing problem-solving and historical knowledge, like unlocking a padlock with the city’s founding date or solving physics puzzles related to factory equipment (Figure 6). These interactive parts make for an immersive experience, merging education and exploration.

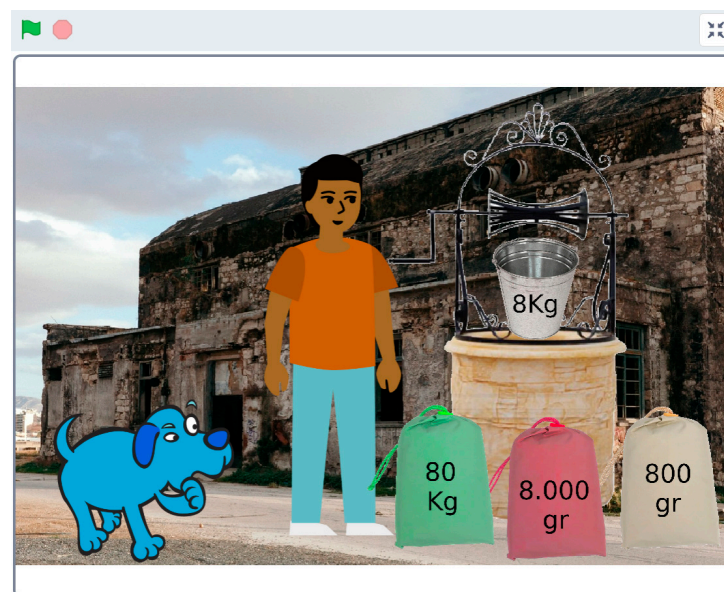


Figure 6. “Math Challenge in an Abandoned Factory”: The player uses math to balance the pulley and lift an 8 kg bucket.

Due to logistical issues (“unfavorable conditions”), the game has not yet had a formal classroom evaluation. However, its designers tested it during development using the AL-MAS framework, measuring engagement, motivation, and learning (see Tables A1–A5 in Appendix A). The plan is to introduce the game soon in local schools to teach Drapetsona’s urban history. Following implementation, student and educator feedback will be gathered to assess its educational impact, engagement levels, and success in fostering historical awareness.

These four case studies (*Super Turtle Adventures*, *Musical Mastermind*, *The Legend of the Dead Knight's Spirit*, and *Urban History of Drapetsona*) showcase practical applications of the design principles discussed earlier. Evaluating their impact on learning, engagement, and user experience necessitates a robust methodological framework. The following section outlines the mixed-methods approach adopted for this research, detailing how established evaluation frameworks (such as SELEAG, ALMAS, LOES-S, and UDL), alongside the HMPI protocol, were employed to systematically analyze these educational games.

2. Materials and Methods

To address the research questions thoroughly, this study utilized a mixed-methods approach, combining quantitative and descriptive (qualitative) data. Central to this approach was the Human–Machine Pair Inspection (HMPI) framework, which facilitates the analysis of how human cognition interacts with game mechanics during gameplay. HMPI was employed in conjunction with other established evaluation tools to provide a comprehensive assessment of the educational games' design and effectiveness. These included

- The Serious Educational Games Evaluation Framework (SELEAG), which assesses learning outcomes, motivation, usability, and feedback mechanisms.
- The Assessment of Learning and Motivation Software (ALMAS), which measures engagement, learning outcomes, and motivational aspects.
- The Learning Object Evaluation Scale for Students (LOES-S), which evaluates digital tools from the student perspective in terms of usability, content quality, and interactivity.
- The Universal Design for Learning (UDL) framework, which promotes accessibility and personalization for diverse learner needs [54].

Combining these different evaluation methods allows for a deeper assessment of how educational games can be improved for maximum impact.

2.1. Qualitative Analysis

The qualitative component of the study—focusing on game design principles, the strengths and limitations of HMPI, and the comparison of different evaluation frameworks—employed a case-study methodology. Participants included small cohorts ($n = 10\text{--}25$) of secondary students (ages 12–16) from three urban schools. All participants and guardians provided informed consent. Ethics approval was secured through institutional review boards aligned with GDPR [55]. Interviews and observations, though qualitative, complemented the quantitative tracking of engagement and learning performance to triangulate findings. This method was chosen to gain both a broad and in-depth understanding of how different educational games perform across dimensions such as engagement, usability, and pedagogical effectiveness. Four educational games were selected for analysis: *Super Turtle Adventures*, *Musical Mastermind*, *The Legend of the Dead Knight's Spirit*, and *Urban History of Drapetsona*. Each game was examined using established frameworks (SELEAG, ALMAS, LOES-S, UDL) alongside the HMPI protocol. As previously mentioned, HMPI focuses on the interplay between player cognition and game mechanics. Utilizing HMPI provided a better understanding of how players interact with the learning environment, respond to real-time feedback, and engage with adaptive features. This deeper perspective informs recommendations for the optimal design and assessment of educational games. It should be noted that the game *Urban History of Drapetsona* was evaluated by a pilot group of educators rather than students. Their expert feedback was instrumental in identifying design limitations and pedagogical affordances within the game's structure.

2.2. Quantitative Analysis

For the quantitative component, numerical data were collected to measure aspects such as player actions, engagement levels, and learning outcomes. Data analysis was

conducted using standard Python tools (<https://www.python.org/> (accessed on 12 July 2025)). The combination of quantitative and qualitative findings yields a solid and detailed understanding pertinent to the research questions, supporting a thorough analysis of educational game design and evaluation [56].

2.3. Evaluation Strategies and Frameworks

To ensure the effectiveness of educational games, a combination of general evaluation strategies and specific, established frameworks is often used throughout the design process. Common evaluation strategies include

- **Pre- and Post-Tests:** Utilized to assess knowledge acquisition and cognitive gains [57].
- **Observational Studies:** Involving direct player observation, video analysis, or game-play recordings to analyze engagement, interaction patterns, and behaviors [58].
- **Surveys and Focus Groups:** Employed to gather qualitative feedback from students and educators for design refinement [59].
- **Embedded Assessment:** Integrating real-time evaluation mechanisms to monitor player performance and provide adaptive feedback seamlessly within gameplay [60].

Alongside these general strategies, a range of established frameworks provides deeper, more structured insights into specific aspects of Game-Based Learning [61]. Prominent examples include

ALMAS (Assessment of Learning and Motivation Software)

Measures engagement, learning outcomes, and motivation in educational games.

To improve the assessment of motivation, the ALMAS (Assessment of Learning and Motivation Software) framework is utilized. ALMAS is an evaluation tool that measures student engagement, learning outcomes, and motivation in educational games [62]. Its innovation lies in its ability to integrate affective factors, such as engagement, with cognitive outcomes like understanding and performance. The assessment of motivation is conducted through a set of key dimensions that include

- **Learning Effectiveness:** The software's ability to improve learning.
- **Mobilization:** The degree to which the game sustains user interest.
- **Engagement:** How the game holds the student's attention during the activity.
- **Long-term Impact:** The effects of the game on students' skills and knowledge over time.
- **Feedback Quality:** The quality and usefulness of the feedback provided.

Specifically, motivation is assessed using pre- and post-game surveys, observational metrics, and self-reporting scales that capture the user's level of interest and persistence in completing tasks.

ECBA (Emerging Technology-based Competence-Based Assessment)

Evaluates how emerging technologies enhance competence development.

This framework evaluates how well games develop specific user competencies using emerging technologies. Dimensions assessed include competence development via technology, performance analysis against objectives, software adaptability, student proficiency with new technologies, and quality of interaction.

GameFlow

Ensures a balance between challenge and enjoyment in educational games.

The goal of GameFlow is to optimize the gaming experience by balancing enjoyment and learning, grounded in flow theory [63]. Player's engagement is maintained by appropriately calibrating the difficulty to ensure that it remains challenging but not discouraging.

Key dimensions that influence the experience include appropriate challenge, player control, and clear objectives. Providing immediate and useful feedback is also a critical factor. Immersion can be achieved through narrative elements or gameplay challenges. Additionally, opportunities for social interaction enhance participation. Clarity in presentation and usability of the interface are equally important. Finally, to maintain flow, balancing player skill with the challenge they face is of critical importance. This subtle balance, after all, is fundamental for keeping players engaged and fully immersed in the flow experience of the game.

LOES-S (Learning Object Evaluation Scale for Students)

Assessing the effectiveness and usability of digital learning tools from the student perspective.

Developed to gather students' perspectives on the effectiveness, design, and usability of digital learning tools, the Learning Object Evaluation Scale for Students (LOES-S) [64] is widely used in adaptive systems and multimedia environments. A key focus of LOES-S is evaluating educational usefulness—specifically, how effectively a resource facilitates knowledge acquisition and understanding. Features like real-time personalization and adaptivity, known to enhance learning outcomes are central to its assessment. Beyond usefulness, LOES-S scrutinizes content quality, emphasizing accuracy and alignment with curriculum standards. It also prioritizes inclusivity and accessibility, evaluating aspects such as intuitive interfaces and compatibility with assistive technologies. Visual design elements, including multimodal interaction options, clear layouts, and simple navigation, are examined for their role in reducing cognitive load and promoting student engagement. Finally, LOES-S evaluates interactivity and adaptability, highlighting the importance of student autonomy and personalized learning paths in modern, learner-centered environments. This learner-centered approach makes LOES-S a highly valuable framework for assessing digital educational experiences.

MEEGA+ (Motivation, Engagement, and Enjoyment in Games Assessment)

Assessing Motivation, Engagement, and Enjoyment in Educational Games.

The MEEGA+ framework evaluates how educational games influence student motivation, engagement, and enjoyment. Motivation refers to the sustained interest students maintain throughout gameplay [65]. Engagement captures the degree of active participation and immersion during the game experience. Enjoyment—understood as the extent to which students find the drill pleasurable and rewarding—is also a central indicator of a game's overall appeal.

SciLOET (Science Learning Object Evaluation Tool)

Evaluates the quality and scientific validity of digital learning objects.

This framework is employed to evaluate digital learning objects specifically within science education [66]. It assesses multiple dimensions to ensure comprehensive quality. The Pedagogical Dimension examines aspects of instructional design and the support provided to learners. Concurrently, the Scientific Dimension focuses on the accuracy of the scientific content and its alignment with inquiry-based learning approaches. Finally, the Technological Dimension evaluates the object's usability and its compatibility across different technological platforms.

SELEAG (Serious Educational Learning Evaluation Assessment Grid)

Assesses interaction, engagement, and goal alignment in serious games.

This framework assesses user interaction, engagement, and the alignment of educational games with learning goals. Its evaluation criteria encompass several key dimensions. Learning outcomes are evaluated to measure the educational effectiveness of the game. The framework also examines motivation, specifically how well the game sustains player

interest. Usability is another critical factor, referring to the game's accessibility and ease of use for diverse users. Finally, the quality, clarity, and usefulness of in-game feedback and assessment mechanisms are assessed [67].

Universal Design for Learning (UDL)

Design for Learner Variability and Inclusion.

Universal Design for Learning (UDL) is an educational framework that promotes accessibility and inclusion by addressing the diverse needs of learners. One of its main principles is providing multiple means of engagement to support motivation, along with presenting content in varied formats to match different learning styles [68]. It also encourages flexible ways for students to demonstrate their understanding. Evidence shows that applying UDL in practice contributes to more inclusive learning environments and helps improve student motivation. As Mygdanis [69] notes, the goal of UDL is to support the creation of adaptable educational contexts capable of meeting a broad range of student needs.

Each framework offers unique insights into learner experience, instructional quality, and technological affordances, making them essential for a holistic evaluation approach. Additionally, recent research has highlighted the importance of embedding ethical foresight into the evaluation of educational technologies. As Efthymiou et al. (2023) [55] argue, educational tools—especially those employing real-time tracking, AI-based feedback, or adaptive learning mechanisms—must be designed with societal responsibility, equity, and inclusivity in mind. Ethical considerations such as data privacy, algorithmic transparency, and the potential for bias in adaptive systems should be factored into the development and evaluation process. By integrating ethical principles into established evaluation frameworks, educational game developers can ensure not only effective learning outcomes but also alignment with responsible and inclusive technological practices.

2.4. Integrating Real-Time Evaluation into Educational Game Design

The Human–Machine Pair Inspection (HMPI) protocol is a formative evaluation methodology that monitors real-time interaction between a user and an adaptive system. It draws upon HCI principles, learning analytics, and cognitive workload theory to analyze how game mechanics affect learner engagement, understanding, and progression [70]. Understanding student interaction within educational games—alongside the cognitive effort required and the system's real-time adaptability—calls for a structured and responsive evaluation method. The HMPI framework effectively addresses these needs. Instead of relying only on post-release evaluations, HMPI emphasizes the value of ongoing feedback throughout the development process. This steady stream of insights enables designers to refine usability, ensure pedagogical value, and maintain player engagement across all stages of game creation.

2.4.1. The Role of HMPI in Educational Game Evaluation

HMPI plays an important role in improving educational games by examining players' behavior and cognitive responses during gameplay. When user feedback is incorporated in real time, it becomes easier to fine-tune game mechanics for deeper immersion and more effective learning experiences. Moreover, this continuous exchange of information also supports the personalization of learning, allowing the game to adjust aspects like difficulty or provide more targeted guidance to individual players. Furthermore, it helps measure how gameplay affects skills like logical reasoning, teamwork, and critical thinking. By checking user interactions continuously, HMPI helps developers ensure their games are engaging, adaptive, and educationally effective throughout the design process, not just at the end.

Key functions of HMPI that help game evaluation:

- Tracking Engagement: Monitors how players interact and how adaptive features affect motivation.
- Addressing Learning Effectiveness: Analyzes thinking effort, problem-solving approaches, and knowledge retention to ensure meaningful learning.
- Evaluating Personalized Experiences: Checks how well AI or other systems adapt game difficulty, feedback, and pacing.
- Measuring Skill Development: Gauges the game's impact on logic, teamwork, and critical thinking.

Additionally, ethical considerations—such as informed consent, data anonymization, and transparency in adaptive feedback systems—were carefully observed during gameplay evaluation. These precautions ensure that real-time data collection and analysis respect student autonomy and privacy in educational contexts.

2.4.2. HMPI Across the Educational Game Development Lifecycle

Integrating HMPI at various stages of the educational software development lifecycle assists in minimizing usability issues proactively and sustaining the effectiveness of adaptive learning models. This approach complements iterative playtesting methodologies, permitting refinements before significant problems escalate.

Application of HMPI is relevant across key development stages:

- Design Phase: Informing interface design and initial usability considerations through HCI principles.
- Development Phase: Refining difficulty scaling and user feedback mechanisms via iterative playtesting.
- Evaluation Phase: Utilizing real-time analytics to assess cognitive workload, engagement levels, and learning efficacy.
- Post-Deployment Analysis: Monitoring player interaction trends after release using adaptive tracking and embedded assessment data.

Figure 7 illustrates how HMPI mechanisms integrate across the four core phases of development—design, implementation, evaluation, and post-deployment—providing actionable data at each stage.



Figure 7. A model for applying Human–Machine Pair Inspection throughout game development.

2.5. Integrating Evaluation Throughout Development

Effective evaluation should be integrated throughout the development process, following a structured model such as the waterfall methodology (Figure 1). Recent implementations in educational game development, such as the work by Helda, Sanusi, and Suhendra [71], demonstrate how the waterfall model facilitates clarity and systematic progress across design phases:

- Early Stages (Design/Requirements): Emphasis on usability heuristics and accessibility standards (e.g., applying UDL guidelines).
- Middle Stages (Implementation/Testing): Direct usability testing, assessment of engagement and motivation (e.g., using SELEAG, ALMAS, LOES-S), analysis of cognitive load, and tracking of learning analytics are critical. Validation of content accuracy (e.g., via SciLOET) and assessment of competence development through technology (e.g., via ECBA) are also relevant here.
- Later Stages (Deployment/Maintenance): Long-term monitoring of engagement, motivation, and enjoyment (e.g., using MEEGA+) provides ongoing insights.

The systematic application of suitable evaluation methods and frameworks at each phase enables continuous refinement, ensuring educational games remain accessible, motivating, and pedagogically sound. Figure 8 maps evaluation tools (e.g., SELEAG, UDL, LOES-S) to the waterfall stages, clarifying when and how to apply them for maximal insight. This alignment helps ensure that evaluation is not isolated but embedded meaningfully throughout the development cycle.

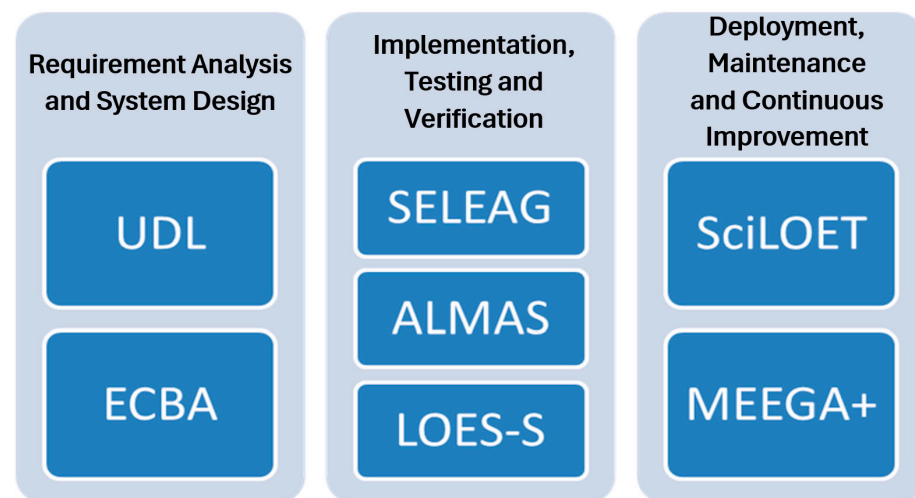


Figure 8. Alignment of evaluation strategies with the phases of educational software development following the waterfall model.

2.6. Advancing HMPI in Educational Game Development

Looking ahead, analytics methodologies derived from HMPI are anticipated to be pivotal in advancing personalized and adaptive learning experiences within educational games. Future innovations may concentrate on AI-based real-time assessments, evaluation models incorporating immersive technologies, and broader cross-disciplinary applications.

Emerging Trends in HMPI-Driven Evaluation Include

- AI-driven Adaptive Learning Analytics: Real-time customization of gameplay based on behavioral data.
- AR and VR for Immersive Learning: Utilizing immersive environments not only for learning delivery but also for assessment.

- Personalized Learning Pathways: Employing AI to adapt challenge progression based on learner performance.
- Cross-Disciplinary Applications: Extending HMPI and similar frameworks to evaluate educational tools in diverse fields like STEAM, humanities, and vocational training.
- Gamified Assessment: Embedding evaluation within authentic, contextualized game challenges.

As these evaluation approaches evolve, they are poised to enhance the design of educational games, making them more personalized, engaging, and pedagogically effective. Notably, models like GEB [72] exemplify the integration of pedagogy with interactive technology.

The synthesis of these diverse evaluation methodologies and frameworks facilitates a comprehensive assessment of educational game performance. This multifaceted approach also facilitates the identification of specific strengths and weaknesses, yielding valuable insights for iterative improvement and future design. Accordingly, the following section details the empirical results of the selected cases obtained from applying these methods.

3. Results

This section presents the findings of the study, detailing the application of the Human–Machine Pair Inspection (HMPI) framework across four educational games. The results are organized around the key research questions, with a focus on interactive game mechanics, engagement patterns, and the effectiveness of HMPI-driven learning models. Table 1 summarizes the core characteristics of each game, including technologies used and human–machine interaction patterns, while Table 2 presents the comparative results of evaluations from multiple frameworks. This approach, which was based on the triangulation of metrics, qualitative feedback, and gameplay data analysis, ensured the reliability and depth in interpreting each game’s educational effectiveness.

Table 1. The game mechanics, evaluation, and Human–Machine Pair Inspection.

Game’s Name	<i>Super Turtle Adventures</i>	<i>Urban History of Drapetsona</i>	<i>Musical Mastermind</i>	<i>The Legend of the Dead Knight’s Spirit</i>
Programming Language	Scratch	Scratch	Micro:bit (Microcode)	Google Forms, Maps and G-Drive objects
Human Interaction	❖ Structured environment to learn Logo	❖ Point-and-click puzzle adventure	❖ Hands-on micro:bit melody decoding	❖ GPS + QR codes, location-based puzzles
Machine Interaction	❖ Intelligent guide	❖ Narrative-driven feedback	❖ Interactive feedback	❖ Real-time geolocation feedback
Evaluation Phases of Construction	SELEAG Design → Maintenance	ALMAS Design → Implementation	UDL Design → Deployment	LOES-S Design → Deployment

Table 2. Comparative evaluation results by framework and game.

Evaluation Method/Game	<i>Super Turtle Adventures</i>	<i>The Legend of the Dead Knight’s Spirit</i>	<i>Musical Mastermind</i>	<i>Urban History of Drapetsona</i>
ALMAS (Motivation)	Moderate—Novelty appreciated, but repetitive tasks reduced motivation.	High—Real-world exploration was highly engaging.	High—Decoding music engaged auditory learners.	Moderate—Educator reviewers noted narrative complexity but appreciated puzzle-based engagement.
SELEAG (Usability and Feedback)	High—Intuitive interface, responsive guide.	Moderate—Some learners struggled with QR interactions.	Moderate—Input validation helpful but not always accurate.	Low—Feedback was inconsistent, and interface usability raised concerns among educators.

Table 2. Cont.

Evaluation Method/Game	<i>Super Turtle Adventures</i>	<i>The Legend of the Dead Knight's Spirit</i>	<i>Musical Mastermind</i>	<i>Urban History of Drapetsona</i>
LOES-S (Student Perspective)	Positive—Students liked structured progression.	Mixed—Appreciated the novelty but wanted clearer instructions.	Positive—Enjoyable physical interaction and sound-based feedback.	-
UDL (Accessibility and Flexibility)	High—Multiple pathways and task supports.	Moderate—GPS-based puzzles limit accessibility.	High—Supported diverse learners with different sensory input.	-

3.1. HMPI Case Study: *The Legend of the Dead Knight's Spirit*

This location-based game, set in the Old Town of Rhodes, combines historical exploration with puzzle-solving. The HMPI analysis showed high motivation from its immersive narrative (ALMAS framework) but moderate usability (SELEAG framework), as some learners struggled with QR code interactions. From the student perspective (LOES-S), results were mixed, with many appreciating the novelty but requesting clearer instructions. The game received a moderate rating for accessibility (UDL framework) because its GPS-based puzzles limited inclusivity.

3.2. HMPI Case Study: *Urban History of Drapetsona*

Developed as a point-and-click puzzle game, this project immerses players in Drapetsona's industrial history. HMPI analysis for this game was derived exclusively from evaluations by the scientific team, as it has not yet undergone full classroom implementation. ALMAS scores were moderate, reflecting engaging puzzles but complex narratives. SELEAG scores were low due to inconsistent feedback and interface usability issues. Neither the LOES-S nor UDL frameworks were applied because the game was not evaluated directly by students.

3.3. HMPI Case Study: *Super Turtle Adventures*

This game provides a structured programming environment for learning Logo through block-based coding. HMPI evaluation showed moderate ALMAS scores—novelty was appreciated, but repetitive tasks slightly reduced motivation. SELEAG results were high, reflecting an intuitive interface and responsive feedback. LOES-S ratings were positive, with students enjoying the structured progression. UDL scores were high, acknowledging the multiple pathways and supports for diverse learning needs.

3.4. HMPI Case Study: *Musical Mastermind*

This game blends music theory and computational thinking using micro:bit devices. The HMPI findings showed high ALMAS motivation scores, as decoding music engages auditory learners. SELEAG scores were moderate, with input validation being helpful but not always accurate. LOES-S results were positive, reflecting enjoyment of the physical interaction and sound-based feedback. UDL evaluation rated the game highly for accommodating diverse learner needs.

3.5. Cross-Case Analysis and Framework Comparison

Looking ahead, analytics methodologies derived from HMPI are anticipated to be pivotal in advancing personalized and adaptive learning experiences within educational games. Future innovations may concentrate on AI-based real-time assessments, evaluation models incorporating immersive technologies, and broader cross-disciplinary approaches.

Across the four case studies, HMPI proved to be a dynamic learning model that fostered interdisciplinary skill development. Key aspects included

- **Interaction and Engagement:** Students interacted with both digital and physical tools (e.g., GPS, QR codes, micro:bit). Games promoting hands-on engagement and exploratory problem-solving lead to deeper participation.
- **Multidisciplinary Skill Development:** The games successfully combine elements of STEAM, history, and language, providing a holistic educational experience.
- **Inclusivity:** The gender analysis indicated that a balance among different gameplay elements (competition, exploration, structured learning) is essential for ensuring inclusivity.
- **The cross-case comparison,** as detailed in Table 2, demonstrates how each evaluation framework provides complementary insights. ALMAS primarily measures motivation, SELEAG assesses usability, LOES-S captures the student experience, and UDL evaluates accessibility and adaptability. By integrating these frameworks, this study provides a more holistic and actionable assessment of each game's effectiveness.

The results from these four case studies demonstrate the adaptability of the HMPI framework across varied educational contexts. They also highlight the added value of integrating multiple evaluation frameworks, as seen in the complementary insights of ALMAS, SELEAG, LOES-S, and UDL. Building on these empirical findings, the next section discusses their broader implications for educational game design and the advancement of human-machine collaborative learning environments. By integrating these frameworks, this study provides a more holistic and actionable assessment of each game's effectiveness [73].

4. Discussion

Building on the results, this section examines how the findings align with educational objectives, evaluates the strengths and limitations of HMPI, and compares multiple evaluation frameworks. It also considers emerging trends in educational game development and offers forward-looking recommendations.

4.1. Emerging Trends in Educational Game Development

As technology advances, several key trends appear to be shaping the future of Game-Based Learning:

- **AI-Driven Adaptive Learning:** A prominent trend involves AI-powered educational games designed to dynamically adjust content, difficulty, and feedback based on player behavior and performance.
- **Immersive Technologies (VR and AR):** Virtual and augmented reality show significant potential to enhance hands-on exploration, making learning more experiential and engaging.
- **Cross-Disciplinary Educational Games:** There is increasing integration of STEAM and humanities within single games, aiming to create more holistic learning experiences [74].
- **Gamification of Assessment:** Traditional assessment methods are progressively incorporating gamification elements, leading to more interactive and potentially more student-driven evaluations.

Human-Machine Pair Inspection (HMPI) Expansion: The application of HMPI is expected to extend beyond educational games into related areas such as virtual classrooms, AI tutors, and interactive simulations.

4.2. Broader Implications of HMPI in Education

Gamified learning environments enriched by human–machine collaboration can increase student motivation and cultivate essential 21st-century skills. By integrating digital tools with educational frameworks, HMPI ensures that learning remains engaging, accessible, and relevant in an increasingly technology-driven world. The comparative analysis of these four games shows the flexibility of HMPI, demonstrating its effectiveness across varied learning contexts (Table 1). Each game is evaluated based on programming language, human–machine interaction (HMI), evaluation frameworks, and development phases, showcasing unique technologies, interaction models, and assessment methodologies designed to create meaningful and engaging learning experiences. This diversity underscores the versatility of HMPI, demonstrating its potential to substantially impact education by making learning more interactive, interdisciplinary, and impactful.

4.3. Limitations of the Study

While this study provides valuable insights into the role of human–machine interaction and adaptive feedback in educational games, it is subject to certain limitations:

- **Sample Size and Scope:** The study analyzed a limited number of case studies, which may not fully capture broader trends in educational game design and effectiveness.
- **Generalizability:** The findings are based on specific Game-Based Learning contexts and technologies, which may not be directly applicable to all educational settings.
- **Methodological Constraints:** Although the study employed both qualitative and quantitative analysis, further refinement of evaluation criteria (e.g., comparative assessments across multiple game types) is necessary to enhance validity.
- **Real-World Testing:** The study's conclusions relied on controlled environments rather than large-scale classroom implementations, requiring further empirical validation with diverse participants.

These limitations are further elaborated in Section 5.1, along with suggestions for overcoming them in future research.

4.4. Addressing Research Question 1: Alignment of Game Design with Educational Objectives

RQ1: How can game design principles and interactive mechanics align effectively with educational objectives to enhance student engagement and learning outcomes?

Game design principles and interactive mechanics must be strategically aligned with educational objectives to create meaningful learning experiences that maximize engagement, knowledge retention, and skill acquisition. A fundamental strategy is to structure challenges so that difficulty gradually increases, allowing learners to build their abilities step by step. Complementing this design approach with adaptive feedback systems provides immediate, personalized responses that reinforce learning and maintain engagement. Additionally, features such as leaderboards, rewards, and time-based challenges help sustain motivation. Research also highlights the value of interactive storytelling, challenge-based progression, and collaborative learning in enhancing conceptual understanding and long-term retention. This was observed in the design of *The Legend of the Dead Knight's Spirit* and *Musical Mastermind*, where the integration of structured challenges with responsive, narrative-driven feedback improved students' conceptual grasp and intrinsic motivation. Ensuring accessibility and adaptability is equally important, meaning that tasks should be offered in multiple formats—visual, auditory, and hands-on—while accommodating varying levels of difficulty. As technology advances, innovations such as AI customization, augmented reality, and immersive storytelling can further align game mechanics with authentic, real-world contexts, enhancing both engagement and learning effectiveness.

4.5. Addressing Research Question 2: Benefits and Limitations of HMPI

RQ2: What are the benefits and limitations of Human–Machine Pair Inspection (HMPI) in evaluating educational games, and how do they influence learning outcomes and player engagement?

A. Key Benefits of HMPI

- **Personalized Learning and Adaptive Feedback:** HMPI supports real-time adaptation of game difficulty and feedback based on user performance, ensuring balanced challenges that sustain engagement and accommodate individual learning needs.
- **Enhanced Learning Outcomes:** Immediate feedback fosters faster knowledge acquisition and retention. Research indicates that students interacting with HMPI-driven educational games exhibit higher retention rates and improved problem-solving skills.
- **Increased Engagement:** The dynamic nature of HMPI enables immersive, interactive learning experiences, keeping students motivated through personalized challenges and continuous progression tracking.
- **Data-Driven Insights for Game Improvement:** Analytics on player behavior, cognitive load, and engagement patterns allow developers and educators to refine game design and optimize learning experiences.
- **Interdisciplinary Skill Development:** Educational games that incorporate HMPI promote cross-disciplinary learning by blending elements of STEAM and humanities, fostering creativity, computational thinking, and critical reasoning [75].

B. Key Limitations of HMPI

- **Data Interpretation Challenges:** HMPI systems produce a large volume of data. Making sense of this data effectively requires a good understanding of both how students learn (educational psychology) and data analysis techniques (data science). If the data are not interpreted correctly, any changes made to educational designs might not help students improve.
- **Scalability and Resource Intensity:** Implementing HMPI requires appropriate technological infrastructure, posing challenges for underfunded institutions.
- **Balancing Automation with Human Input:** While automated feedback from HMPI can be a great help to learners, relying too much on these automated systems might reduce important opportunities for human interaction. This includes valuable moments like discussions led by teachers, students working together, and direct guidance from educators.
- **Connecting with Teaching Strategies:** HMPI primarily looks at how a learner interacts with a specific system or game. For HMPI to be truly effective in an educational setting, it must align well with broader teaching methods and overall educational goals. Its success depends on how seamlessly it fits into the existing pedagogical framework.

C. Impact of HMPI on Learning Outcomes and Player Engagement

- **Improved Knowledge Retention:** Personalized feedback and adaptive difficulty levels help students retain concepts more effectively.
- **Higher Motivation and Immersion:** Real-time interaction and progress tracking maintain student interest.
- **Balanced Learning Experience:** HMPI's mix of challenge and support prevents frustration or disengagement.
- **Inclusive Game Design:** Analyzing gameplay data enables targeted refinements to address diverse cognitive abilities and engagement preferences.

In conclusion, while HMPI offers substantial advantages in enhancing learning experiences and engagement in educational games, its overall success relies on careful implementation, balanced integration with human instruction, and accurate interpretation of data.

4.6. Evaluating Educational Games Through Multiple Frameworks

RQ3: In what ways do different evaluation frameworks—such as ALMAS, ECBA, Game-Flow, and SELEAG—differ in how they measure the effectiveness of educational games in terms of player motivation, cognitive skill development, and knowledge retention?

Assessing the effectiveness of educational games requires multidimensional frameworks that can capture the complexity of learning experiences. Each framework offers distinct lenses through which to evaluate core components such as engagement, skill acquisition, and retention [76]. A comparative summary of evaluation results across the four games examined is provided in Table 2, which illustrates how each framework highlights different strengths and weaknesses per game.

Based on the application of the different frameworks, several key insights were gained. The ALMAS framework, which focuses on motivation and long-term engagement showed that *The Legend of the Dead Knight's Spirit* and *Musical Mastermind* were highly rated due to their engaging narrative and real-world exploration. *Super Turtle Adventures* received a moderate score because repetitive tasks reduced motivation, while *Urban History of Drapetsona* also scored moderately due to narrative complexity noted by educator reviewers. The SELEAG framework, which concentrates on usability and pedagogical alignment, found *Super Turtle Adventures* to be highly effective due to its intuitive interface. *Urban History of Drapetsona* received a low score because of inconsistent system feedback and interface issues. *The Legend of the Dead Knight's Spirit* and *Musical Mastermind* were rated as moderate due to some learners struggling with QR interactions and inconsistent input validation, respectively.

The LOES-S framework, which reflects the student perspective highlighted that both *Super Turtle Adventures* and *Musical Mastermind* were positively received for their structured progression and enjoyable interaction. *The Legend of the Dead Knight's Spirit* received mixed results, with students appreciating the novelty but requesting clearer instructions. LOES-S was not applied to *Urban History of Drapetsona*, as it was not evaluated by students. Finally, the UDL framework, which promotes accessibility and personalization, rated *Super Turtle Adventures* and *Musical Mastermind* highly for supporting diverse learner needs. *The Legend of the Dead Knight's Spirit* scored moderately, as its GPS-based puzzles limited accessibility. UDL was not applied to *Urban History of Drapetsona* for the same reason as LOES-S.

4.7. Synthesizing Insights and Looking Forward

The synthesis of the findings reveals a dynamic interplay among game mechanics, pedagogical goals, learner engagement, and evaluative methodologies. At the center of this ecosystem lies the principle of alignment, ensuring that game mechanics meaningfully correspond to intended learning outcomes. This alignment becomes more powerful when it is supported by adaptive feedback systems, such as those utilized within the HMPI framework. As shown in previous sections, such systems enhance personalization, sustain engagement, and promote deeper learning. The HMPI approach offers valuable contributions to educational game evaluation, particularly in its ability to generate actionable data on learner behavior and performance. However, as the findings suggest, the successful implementation of HMPI is contingent upon a few factors. These include the capacity to interpret complex data outputs, the availability of institutional resources, and the alignment of automated systems with broader pedagogical strategies. In addition, learner diversity

remains a critical consideration. Variations in engagement across demographic lines, such as gender or cognitive style, highlight the need for inclusive and adaptable game design that supports equity in learning experience.

Moreover, the evaluation frameworks discussed in Section 4.3 reinforce the importance of multidimensional assessment. Tools such as ALMAS, ECBA, GameFlow, and SELEAG offer complementary strengths, and their combined use allows for more comprehensive and context-sensitive analysis. The synergy between HMPI and these frameworks presents a compelling direction for future research and practice. Moving forward, efforts should concentrate on refining the integration of HMPI and hybrid evaluation models across diverse educational settings. This includes testing scalability, optimizing real-time analytics, and supporting educators with practical tools for interpretation and implementation. Technological innovation, particularly in areas such as AI-driven feedback, adaptive storytelling, and multimodal interaction—can further enhance the alignment between game mechanics and authentic learning needs. Ultimately, the findings underscore a critical insight: the effectiveness of serious games depends not only on innovative design or powerful technologies, but on how these elements are meaningfully combined, evaluated, and adapted to diverse learner populations.

5. Conclusions

Rigorous evaluation of educational games is essential for validating their pedagogical relevance, ensuring high design quality, and confirming their sustained impact on student learning. The integration of structured evaluation models—such as SELEAG, GameFlow, and ECBA—equips educators and developers with essential tools to understand what drives learner motivation, fosters engagement, and facilitates cognitive development. These frameworks provide both analytic rigor and design-oriented insights, enabling the creation of personalized and inclusive learning experiences that respond to diverse learner profiles. A central contribution of this study lies in the demonstrated potential of the HMPI approach in strengthening the educational effectiveness of serious games. HMPI supports critical components of Game-Based Learning, including adaptive feedback, interactive storytelling, and cognitively engaging activities. Its interdisciplinary foundation—linking STEAM, humanities, and cognitive science—positions it as a powerful mechanism for promoting creativity, problem-solving, and collaborative learning.

Moreover, HMPI-generated insights offer meaningful support for inclusive instructional design, highlighting the importance of addressing individual learning differences and gender-based engagement patterns [77]. The combination of gamification and artificial intelligence for personalizing gameplay has also shown promising results, positively influencing academic achievement, learner motivation, and depth of engagement [78]. In addition, the implementation of multimodal learning strategies—such as integrating hands-on experimentation with problem-solving scenarios—has been found to enhance memory retention and critical thinking by activating multiple sensory and cognitive channels [79]. These approaches make learning more inclusive and impactful, especially when applied across different educational levels. Advancing the field of educational game design thus requires a balanced integration of technological innovation, pedagogical insight, and inclusive design principles. The findings of this study form a solid foundation for the ongoing development of serious games that are not only engaging and adaptive but also pedagogically robust and socially responsive.

5.1. Challenges and Areas for Further Research

While HMPI and AI-driven educational game models offer significant advantages, several challenges must be addressed to optimize their effectiveness and scalability.

5.1.1. Interpreting Data from HMPI and AI-Driven Feedback

One of the most pressing challenges in the deployment of HMPI and AI-powered educational games is the effective interpretation of the big data that they generate. These systems produce continuous streams of learners' behavioral data feeds, ranging from interaction patterns and response times to cognitive load indicators and engagement metrics. While data feeds hold considerable promise for informing personalized instruction, their utility is contingent on the availability of robust analytic frameworks and user-friendly visualization tools. Educators and developers often lack the technical expertise or infrastructure required to transform raw data into actionable insights. Without advanced analytics and clearly defined pedagogical models for interpretation, there is a risk of misalignment between data-driven recommendations and actual learner needs. In such cases, adaptive feedback mechanisms may become superficial or even counterproductive, failing to support meaningful learning progress. Furthermore, ethical considerations surrounding data privacy, consent, and algorithmic bias must also be considered, particularly when systems are used in formal educational settings. Future research should explore how data from HMPI systems can be meaningfully contextualized within broader educational goals and used to support teachers rather than replace them.

5.1.2. Scalability of Personalized Learning Approaches

As previously demonstrated using HMPI, personalized learning strategies have the potential to enhance student engagement and adaptation. However, implementing such strategies across many students without compromising quality demands substantial technological and infrastructural support. While AI has the potential to tailor learning experiences to individual needs, the computational resources required to maintain real-time adaptation can be costly and logistically demanding. Ensuring that these technologies remain accessible to a broad range of educational institutions, including those with limited resources, is essential for promoting equitable learning opportunities. Future research should explore cost-effective cloud-based solutions and optimized AI algorithms that can facilitate scalability while maintaining high levels of personalization.

5.1.3. Balancing Automation with Human-Led Instruction

As previously demonstrated in the context of HMPI-based systems, AI-driven learning environments enhance adaptability and responsiveness. However, over-reliance on automation may inadvertently diminish the role of human-led discussions and instructor intervention. Personalized instruction from educators plays a crucial role in providing emotional support, fostering critical thinking, and encouraging deeper engagement—elements that AI alone cannot fully replicate. Striking the right balance between automated feedback and teacher involvement is essential to maintaining a meaningful and interactive learning experience. Educational game developers should prioritize designs that complement human instruction rather than replace it, ensuring that AI serves as an enhancement rather than a substitute for traditional teaching methods [80].

5.1.4. Need for Large-Scale Empirical Validation

Controlled case studies have yielded valuable insights into adaptive learning mechanics and Human–Machine Pair Inspection (HMPI), yet they often fail to capture the full complexity of real-world classrooms. To determine how these systems perform in everyday educational settings, large-scale trials are required—engaging more students across a variety of schools and learning levels. Future investigations ought to compare diverse learner populations across multiple institutions and track their progress over time [81]. Such expansive experimentation will reveal how cultural background, infrastructure, and

individual learning differences shape the effectiveness of AI-driven, game-based interventions. While pilot studies offer important initial evidence, mixed-methods replications and comparative field trials are essential to determine whether early findings are held on a scale. Anchoring these efforts in established frameworks—such as Juul’s rule-based formal system model or Sweetser and Wyeth’s GameFlow model—will ensure methodological rigor without sacrificing contextual relevance.

Building this broader empirical foundation would refine evaluation methodologies, strengthen adaptive feedback algorithms, and ensure that AI-driven personalization supports rather than replaces human instruction. Ultimately, this groundwork will enable the development of educational games that are inclusive, scalable, and demonstrably effective at fostering deep learning, skill acquisition, and equitable access.

The following section outlines practical recommendations and future directions, addressing the opportunities and challenges that lie ahead.

6. Recommendations and Future Directions

As previously discussed in the results and discussion sections, adaptive learning systems have shown strong potential for enhancing personalization and engagement. The ongoing evolution of educational games is closely linked to advancements in data analytics, learner-centered design, and emerging technologies. By integrating artificial intelligence AI-driven learning, immersive narrative structures, and evaluation frameworks such as HMPI, educators and developers can design learning experiences that are engaging, inclusive, and pedagogically effective. When coupled with real-world applications, interdisciplinary content, and adaptive learning models, game-based education has the potential to remain a vital and transformative component of contemporary learning environments [82]. To maximize both impact and scalability, future research and development should focus on the following key areas.

6.1. Advancing Adaptive Learning and Personalization

A primary avenue for enhancing educational games lies in the development of AI-powered adaptive learning systems capable of dynamically adjusting difficulty levels in response to learner performance. By analyzing in-game behavior, machine-learning algorithms can identify areas of cognitive struggle and modify challenges in real time to support individual learning trajectories.

Additionally, the implementation of personalized learning pathways allows students to progress at their own pace, engaging with content that aligns with their cognitive styles and preferences. Predictive analytics can further refine game mechanics, ensuring that content remains appropriately scaffolded to sustain motivation and improve knowledge retention.

6.2. Enhancing Interactivity Through Hands-On and Analytical Engagement

Effective educational games should offer a balanced combination of hands-on activities and structured, logic-based problem-solving tasks to address a broad range of learning styles. The inclusion of physical tools—such as micro:bit devices, QR codes, and IoT-enabled resources—enhances the learning experience for kinesthetic learners by linking digital content with tangible interaction. At the same time, strategically designed analytical challenges can stimulate critical thinking and foster deep cognitive engagement. Future developments in Game-Based Learning are expected to integrate physical and digital modalities seamlessly, creating immersive, hybrid environments that foster both practical and analytical learning [83].

6.3. Expanding Gamified Assessments and Alternative Evaluation Methods

As educational games continue to evolve, assessment methodologies should also be adapted to the games' design. Traditional exams and standardized tests are often limited in capturing the depth and dynamism of Game-Based Learning. On the other hand, future assessment systems should incorporate interactive elements such as simulations, scenario-based challenges, and digital storytelling [84].

Embedding real-time performance analytics within games allows for continuous monitoring of cognitive development, engagement levels, and content mastery. AI-driven adaptive quizzes can further personalize assessment by adjusting difficulty and content based on individual learner profiles. They transform evaluation into a fluid and formative component of the learning process.

6.4. Immersive Learning with Augmented Reality (AR) and Virtual Reality (VR)

The integration of AR and VR technologies into educational games offers transformative possibilities for experiential learning [85]. Augmented reality can be used to create location-based storytelling experiences, where students explore historical sites, scientific phenomena, or cultural artifacts through interactive AR overlays. Virtual reality, on the other hand, can provide fully immersive collaborative problem-solving scenarios, enabling students to engage in realistic team-based challenges while developing communication, cooperation, and critical thinking skills. As these technologies become more and more accessible, their application in educational games will provide students with deeper, more interactive learning experiences.

6.5. Scaling up Human–Machine Pair Inspection (HMPI) in Educational Research

The HMPI framework should be expanded to incorporate AI tutors, voice recognition, and multimodal learning environments, further refining interactive educational experiences. As HMPI collects and analyzes data on student interactions, its effectiveness can be enhanced through AI-driven insights that monitor cognitive load, engagement patterns, and learning efficiency. Future research should also explore the integration of emotion recognition technologies to assess affective states such as frustration or confusion, enabling real-time adaptation of content based on learners' emotional and cognitive needs.

6.6. Strengthening Teacher Training and Curriculum Integration

For Game-Based Learning to be effectively implemented, educators must be adequately trained and supported. The successful implementation of Game-Based Learning hinges on strengthening teacher training and facilitating curriculum integration. To this end, professional development programs must equip educators not only with technical skills but also with pedagogical strategies to integrate educational games into their lesson plans and align them with specific curriculum goals. AI-powered tools can further support educators by offering content customization and differentiated instruction based on student data. Additionally, curricular models must evolve to embed Game-Based Learning principles at a structural level, ensuring that games function as core pedagogical instruments rather than as optional enhancements.

6.7. Ensuring Accessibility, Inclusivity, and Bridging the Digital Divide

Universal Design principles must underpin the development of educational games to ensure accessibility for learners of all abilities, including students with learning disabilities. In parallel, developers and policymakers must address the digital divide by expanding access to educational games in underserved regions and resource-constrained schools. Research should prioritize cost-effective, cloud-based platforms and mobile-friendly tech-

nologies to make Game-Based Learning accessible on a global scale. The promotion of open-access development tools will further democratize educational innovation, enabling learners from all socioeconomic backgrounds to benefit.

6.8. Final Thoughts

The application of rigorous evaluation strategies in educational game design holds significant potential to enhance student engagement, motivation, and interdisciplinary learning. These strategies also address critical challenges related to accessibility, personalization, and assessment. Emerging technologies—such as AI, AR/VR, and gamified assessment tools—are rapidly reshaping the educational landscape, enabling learning experiences that are more immersive, adaptive, and interactive. These advancements have shown measurable improvements in learner engagement, motivation, and learning outcomes, paving the way for the next generation of dynamic, impactful Game-Based Learning environments. As technological innovation accelerates, Game-Based Learning is becoming a pivotal medium for engaging students with digital content in ways that are both personalized and inclusive. Ensuring accessibility, ethical AI implementation, and interdisciplinary integration will be crucial in scaling these innovations for global educational impact. The future of education will be driven by intelligent, interactive, and inclusive learning environments, where games serve as powerful tools for knowledge acquisition, skill development, and creative problem-solving.

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Abbreviations

The following abbreviations are used in this manuscript.

Abbreviation	Full Term	Description
AI	Artificial Intelligence	Use of machines and algorithms to simulate human intelligence processes.
ALMAS	Assessment of Learning and Motivation Software	Framework for measuring engagement, motivation, and learning outcomes in educational games.
AR	Augmented Reality	Technology overlaying digital content on the real-world environment.
ECBA	Emerging Technology-based Competence-Based Assessment	Framework assessing competence development using emerging technologies.
GBL	Game-Based Learning	Use of complete games aligned with learning objectives to enhance engagement and understanding.
HCI	Human-Computer Interaction	Field focusing on the design and use of computer technology and user interfaces.
HMPI	Human-Machine Pair Inspection	Protocol for real-time analysis of learner interaction, engagement, and adaptive feedback in educational games.
ICT	Information and Communication Technologies	Technologies used for communication, creation, and sharing of information in education and other fields.
IoT	Internet of Things	A network of interconnected devices that collect and exchange data.
LOES-S	Learning Object Evaluation Scale for Students	A tool for evaluating the usability, content quality, and interactivity of digital learning resources from the student perspective.
MEEGA+	Motivation, Engagement, and Enjoyment in Games Assessment	Framework measuring motivational and engagement aspects of educational games.
SciLOET	Science Learning Object Evaluation Tool	A tool for assessing scientific accuracy, pedagogical design, and technological usability of science-focused learning objects.
SELEAG	Serious Educational Games Evaluation Framework	Evaluation grid focusing on learning outcomes, motivation, usability, and feedback.
STEAM	Science, Technology, Engineering, Arts, and Mathematics	Educational approach integrating arts into STEM for creativity and innovation.
UDL	Universal Design for Learning	Framework for inclusive education that addresses learner variability through multiple means of engagement, representation, and expression.
VR	Virtual Reality	An immersive computer-generated environment simulating real or imagined experiences.

Appendix A

Table A1. Learning outcomes and knowledge acquisition ¹.

Game	Achievement of Learning Objectives (%)	Evaluation Method
<i>Super Turtle Adventures</i>	85%	SELEAG
<i>Musical Mastermind</i>	87%	UDL
<i>The Legend of the Dead Knight's Spirit</i>	88%	LOES-S
<i>Urban History of Drapetsona</i>	84%	ALMAS ²

¹ The effectiveness of the games in promoting learning objectives was assessed based on students' ability to grasp and apply new concepts. ² The assessment was conducted by the designers during the development and implementation phase of the game.

Table A2. Engagement and player experience ¹.

Game	Engagement (Boys) (%)	Engagement (Girls) (%)	Overall Engagement (%)
<i>Super Turtle Adv.</i>	88	89	88.5
<i>Musical Mastermind</i>	90	91	90.5
<i>The Legend of the Dead Knight's Spirit</i>	92	93	92.5
<i>Drapetsona Cult. Game</i>	87	88	87.5

¹ Engagement levels were measured through player participation, motivation, and interaction with game mechanics.

Table A3. Gameplay experience and scenario effectiveness ¹.

Game	Gameplay Quality (%)	Scenario Quality (%)
<i>Super Turtle Adventures</i>	86	89
<i>Musical Mastermind</i>	89	91
<i>The Legend of the Dead Knight's Spirit</i>	91	92
<i>Urban History of Drapetsona</i>	85	88

¹ Each game was assessed based on game mechanics, player interaction, and overall scenario design.

Table A4. Gender-based gameplay preferences ¹.

Game	Boys' Preferred Element	Girls' Preferred Element
<i>Musical Mastermind</i>	Hands-on micro:bit experimentation	Puzzle-solving and melody decoding
<i>The Legend of the Dead Knight's Spirit</i>	Searching for hotspots (competition)	Historical learning and exploration

¹ The research results were based on student questionnaires completed after their gameplay experience.

Table A5. Overall effectiveness ¹.

Game	Overall Effectiveness (%)
<i>Super Turtle Adventures</i>	86
<i>Musical Mastermind</i>	90
<i>The Legend of the Dead Knight's Spirit</i>	92
<i>Urban History of Drapetsona</i>	85
Average	88.25

¹ The research results were based on student questionnaires completed after their gameplay experience.

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